Triple-Higgs production as a probe of the Higgs potential

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Abstract

We study triple-Higgs production in the κ-framework as a direct probe of the Higgs potential. We derive compact parametrisations of the inclusive gluon-fusion production cross section that capture the dependence on the Higgs trilinear and quartic self-couplings together with anomalous two-top–two-Higgs and two-top–three-Higgs interactions. These results provide a practical framework for interpreting current and future LHC measurements. We use the parametrisations to assess the sensitivity of LHC Run 2 and HL-LHC data to Higgs self-couplings and show that differential kinematic distributions can provide additional discrimination between scenarios with nearly identical inclusive production rates.

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1 Introduction

The discovery of the Higgs boson in 2012 [1, 2] completed the particle content of the Standard Model (SM) and established the mechanism of electroweak (EW) symmetry breaking. Since then, a major objective of the LHC physics programme has been to determine the properties of the Higgs boson with increasing precision. While its couplings to EW gauge bosons and third-generation fermions are already constrained at the few- to ten-percent level, its self-interactions remain largely unexplored. Measuring the Higgs self-couplings is therefore one of the most important outstanding goals of collider physics, since they directly probe the shape of the Higgs potential and the dynamics of EW symmetry breaking.

In the SM, the Higgs potential is determined by a single quartic coupling $\lambda = m_h^2/(2v^2)$. After EW symmetry breaking, it predicts unique values for the Higgs trilinear and quartic self-couplings,

$$\lambda_3^{\rm SM} = \lambda\,, \qquad \lambda_4^{\rm SM} = \frac{\lambda}{4}\,, \tag{1}$$

once the Higgs boson mass $m_h \simeq 125\,$GeV and vacuum expectation value $v \simeq 246\,$GeV are fixed. Any deviation from these predictions would provide evidence for physics beyond the SM, such as extended scalar sectors, composite Higgs models, or mechanisms for EW baryogenesis. Reviews of such beyond-the-SM (BSM) scenarios can be found, for example, in Refs. [3, 4].

At the LHC, double-Higgs production provides the primary direct probe of the Higgs trilinear self-coupling, while loop-induced effects in single-Higgs processes yield complementary constraints [5–13]. These measurements are typically interpreted in the κ-framework [14–16], which parametrises deviations from the SM through

$$\kappa_3 = \frac{\lambda_3}{\lambda_3^{\rm SM}}\,, \tag{2}$$

assuming all other couplings take their SM values. Using the full $\sqrt{s} = 13\,$TeV LHC Run 2 data set, the ATLAS and CMS collaborations obtain the combined constraint

$$-0.71 < \kappa_3 < 6.1 \quad (\text{ATLAS} + \text{CMS}) \tag{3}$$

at the 95% confidence level (CL) [17]. The HL-LHC, with an integrated luminosity of $6\,\text{ab}^{-1}$, is expected to improve this sensitivity to

$$0.5 < \kappa_3 < 1.7 \quad (\text{HL-LHC})\,. \tag{4}$$

Constraining the quartic Higgs self-coupling is considerably more challenging. It is conventionally parametrised as

$$\kappa_4 = \frac{\lambda_4}{\lambda_4^{\rm SM}}\,, \tag{5}$$

and triple-Higgs production provides its only direct probe at hadron colliders. However, the tiny SM cross section of approximately 100 ab at $\sqrt{s} = 14\,\text{TeV}$, calculated at next-to-next-to-leading order (NNLO) in QCD [18], renders such measurements experimentally very challenging. Current ATLAS [19] and CMS [20] searches yield

$$-230 < \kappa_4 < 240 \quad (\text{ATLAS})\,, \qquad -190 < \kappa_4 < 190 \quad (\text{CMS})\,, \tag{6}$$

at the 95% CL, assuming all other couplings are SM-like, in particular $\kappa_3 = 1$. A recent HL-LHC study [21] projects these bounds to improve to

$$-81 < \kappa_4 < 89 \quad (\text{HL-LHC}) \tag{7}$$

with an integrated luminosity of $6\,\text{ab}^{-1}$. In addition to triple-Higgs production, double-Higgs production is also sensitive to κ_4 through two-loop EW corrections [21–24], although these constraints are generally weaker.

In this note, we present a comprehensive study of triple-Higgs production within the κ-framework. We derive compact parametrisations of the inclusive gluon-fusion production cross section that capture its dependence on the Higgs trilinear and quartic self-couplings, the top-quark Yukawa coupling modifier κ_t, and the two-top–two-Higgs and two-top–three-Higgs interactions κ_{2t} and κ_{3t}. These expressions provide practical tools for interpreting current and future LHC measurements. Using them, we investigate the sensitivity of current and projected LHC data to the Higgs self-couplings and study the impact of correlated modifications of the top-Higgs interactions. We further demonstrate that differential kinematic distributions retain significant sensitivity even in scenarios with nearly degenerate inclusive cross sections, highlighting the importance of shape information in future triple-Higgs searches.

2 Triple-Higgs production in the κ-framework

Triple-Higgs production at hadron colliders has been the subject of extensive theoretical investigation over the past two decades [18, 21, 22, 25–41]. Within the SM, the gluon-fusion process $gg \to 3h$ is dominated by the one-loop topologies shown in the upper and middle rows of Figure 1. In the figure, green (blue) boxes denote insertions of the Higgs trilinear (quartic) self-coupling, while orange circles represent the top-quark Yukawa coupling. The two diagrams in the lower row illustrate representative BSM contributions involving a contact interaction between two top quarks and two Higgs bosons. In the SM, bottom-quark loops modify the $gg \to 3h$ production cross section by only about 0.2% at LHC energies [39]; they are nevertheless retained in the numerical calculations below.

The κ-formalism Lagrangian, which parametrises the interaction vertices highlighted in Figure 1, takes the form

$$\mathcal{L}_\kappa \supset -\kappa_3 v \lambda h^3 - \kappa_4 \frac{\lambda}{4} h^4 - \kappa_t \frac{y_t}{\sqrt{2}} \bar{t}th - \kappa_{2t} \frac{y_t}{\sqrt{2}v} \bar{t}th^2 - \kappa_{3t} \frac{y_t}{2\sqrt{2}v^2} \bar{t}th^3\,, \tag{8}$$

where κ_3 and κ_4 were introduced in Eqs. (2) and (5), respectively, and $y_t = \sqrt{2}m_t/v$, with m_t denoting the top-quark mass. The parameter κ_t thus encodes modifications of the top-quark Yukawa coupling, while κ_{2t} and κ_{3t} parametrise contact interactions between two top quarks and two or three Higgs bosons, respectively. The final term can equivalently be written

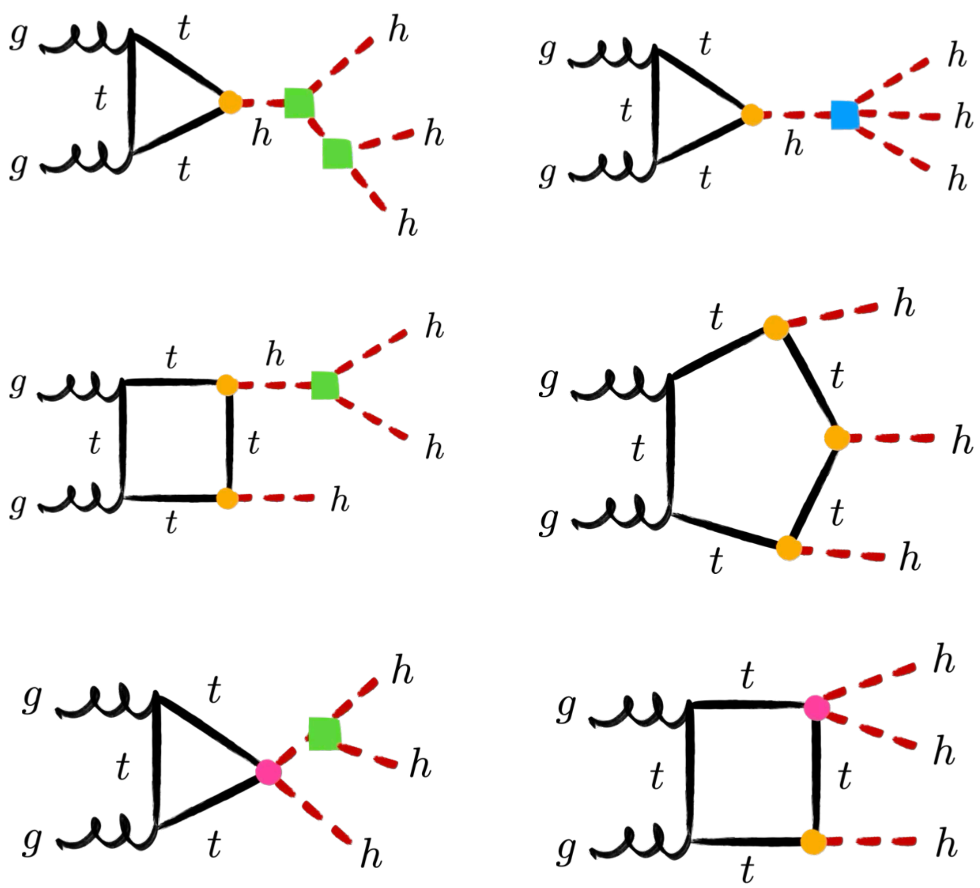

Figure 1: Representative one-loop Feynman diagrams contributing to triple-Higgs production via gluon fusion, illustrating the triangle, box, and pentagon topologies. Green (blue) boxes denote the Higgs trilinear (quartic) self-coupling, while orange (magenta) circles indicate top-quark couplings involving one (two) Higgs bosons.

as $-\kappa_{3t} m_t \bar{t} t h^3/(2v^3)$ [35]. The SM is recovered from Eq. (8) for $\kappa_3 = \kappa_4 = \kappa_t = 1$ and $\kappa_{2t} = \kappa_{3t} = 0$. In the UFO implementation used below, the self-coupling inputs are $D_3 = \kappa_3 - 1$ and $D_4 = \kappa_4 - 1$; consequently, $D_3 = D_4 = 0$ corresponds to $\kappa_3 = \kappa_4 = 1$, not to vanishing physical self-couplings.

To determine the dependence of the inclusive $gg \to 3h$ production cross section on κ_3, κ_4, κ_t, and κ_{2t}, we employ the leading-order (LO) matrix elements calculated in Ref. [39] and implemented in the `MCFM` [42–45] parton-level event generator. We use $m_h = 125\,\text{GeV}$, $m_t = 173\,\text{GeV}$, and $m_b = 4.77\,\text{GeV}$ as input parameters together with the `PDF4LHC21` parton distribution functions (PDFs) [46]. The renormalisation and factorisation scales, μ_R and μ_F, are chosen dynamically as $\mu_R = \mu_F = m_{3h}/2$, where m_{3h} denotes the invariant mass of the three-Higgs system. For collision energies of $\sqrt{s} = 13\,\text{TeV}$, $\sqrt{s} = 13.6\,\text{TeV}$, and $\sqrt{s} = 14\,\text{TeV}$, we parametrise the LO cross section as

$$\sigma_{\text{LO}}^{\sqrt{s}} = \left(\sigma_{\text{LO}}^{\sqrt{s}}\right)_{\text{SM}} \sum_{n,m} c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t}) \left(\Delta\kappa_3\right)^n \left(\Delta\kappa_4\right)^m , \tag{9}$$

where $\Delta\kappa_i = \kappa_i - 1$ for $i = 3, 4$. The corresponding LO SM cross sections are

$$\sigma_{\text{LO}}^{13\,\text{TeV}} = 42.1\,\text{ab}, \qquad \sigma_{\text{LO}}^{13.6\,\text{TeV}} = 47.9\,\text{ab}, \qquad \sigma_{\text{LO}}^{14\,\text{TeV}} = 51.0\,\text{ab}. \tag{10}$$

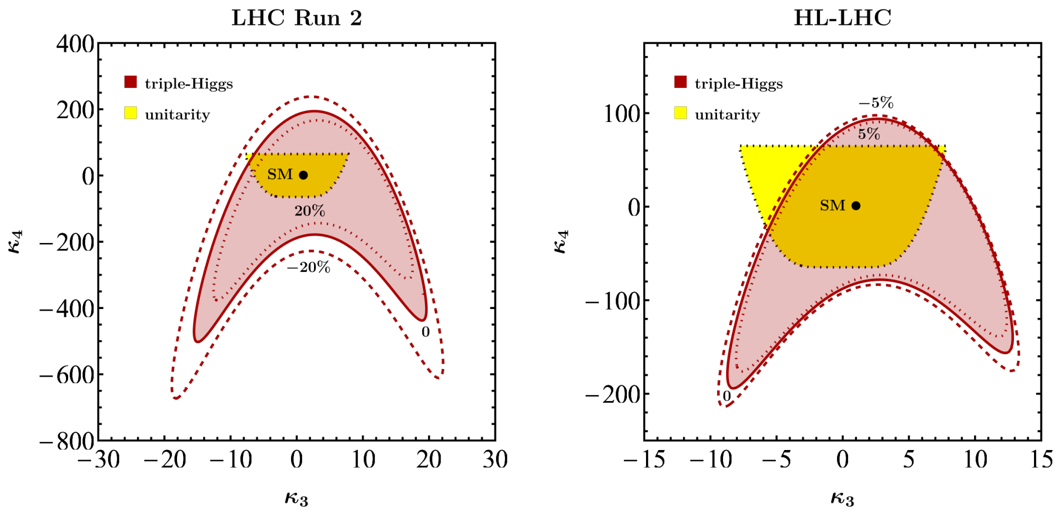

Figure 2: Constraints in the κ_3–κ_4 plane at LHC Run 2 (left) and the HL-LHC (right). The red contours indicate the preferred 95% CL regions obtained from triple-Higgs production in gluon fusion for three representative values of κ_t, as specified by the contour labels. We assume $\kappa_{2t} = 0$. The SM prediction is denoted by black points. The yellow regions, enclosed by black dotted contours, represent the perturbative unitarity bounds derived from tree-level $hh \to hh$ scattering.

For $\sqrt{s} = 13\,\text{TeV}$, the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ appearing in Eq. (9) are

$$\begin{aligned} c_{00}^{13\,\text{TeV}}(\kappa_t, \kappa_{2t}) = &-6.05 \cdot 10^{-4} + 0.0384\kappa_t^2 - 0.350\kappa_t^3 + 1.47\kappa_t^4 - 3.04\kappa_t^5 + 2.89\kappa_t^6 \\ &+ \kappa_{2t}\left(1.61\kappa_t^2 - 7.82\kappa_t^3 + 15.6\kappa_t^4 - 14.8\kappa_t^5\right) \\ &+ \kappa_{2t}^2\left(6.96 - 22.3\kappa_t + 37.0\kappa_t^2\right), \end{aligned} \tag{11}$$

$$c_{01}^{13\,\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0829\kappa_t^2 - 0.264\kappa_t^3 + 0.0895\kappa_t^4 + \kappa_{2t}\left(0.673\kappa_t - 1.12\kappa_t^2\right), \tag{12}$$

$$c_{02}^{13\,\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0169\kappa_t^2, \tag{13}$$

$$\begin{aligned} c_{10}^{13\,\text{TeV}}(\kappa_t, \kappa_{2t}) = &\,0.211\kappa_t^2 - 1.45\kappa_t^3 + 3.65\kappa_t^4 - 3.28\kappa_t^5 \\ &+ \kappa_{2t}\left(3.49\kappa_t - 13.4\kappa_t^2 + 15.6\kappa_t^3\right) + \kappa_{2t}^2\left(13.9 - 22.3\kappa_t\right) \end{aligned} \tag{14}$$

$$c_{11}^{13\,\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0978\kappa_t^2 - 0.264\kappa_t^3 + 0.673\kappa_{2t}\kappa_t, \tag{15}$$

$$c_{20}^{13\,\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.288\kappa_t^2 - 1.25\kappa_t^3 + 1.84\kappa_t^4 + \kappa_{2t}\left(2.81\kappa_t - 6.70\kappa_t^2\right) + 6.96\kappa_{2t}^2, \tag{16}$$

$$c_{21}^{13\,\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0489\kappa_t^2 \tag{17}$$

$$c_{30}^{13\,\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.159\kappa_t^2 - 0.417\kappa_t^3 + 0.938\kappa_{2t}\kappa_t, \tag{18}$$

$$c_{40}^{13\,\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0396\kappa_t^2. \tag{19}$$

At $\sqrt{s} = 13.6\,\text{TeV}$, the corresponding coefficients read

$$\begin{aligned} c_{00}^{13.6\,\text{TeV}}(\kappa_t, \kappa_{2t}) = &-9.34 \cdot 10^{-3} + 0.0572\kappa_t^2 - 0.446\kappa_t^3 + 1.68\kappa_t^4 - 3.17\kappa_t^5 + 2.89\kappa_t^6 \\ &+ \kappa_{2t}\left(1.59\kappa_t^2 - 7.69\kappa_t^3 + 15.3\kappa_t^4 - 14.6\kappa_t^5\right) \\ &+ \kappa_{2t}^2\left(6.89 - 22.1\kappa_t + 36.8\kappa_t^2\right), \end{aligned} \tag{20}$$

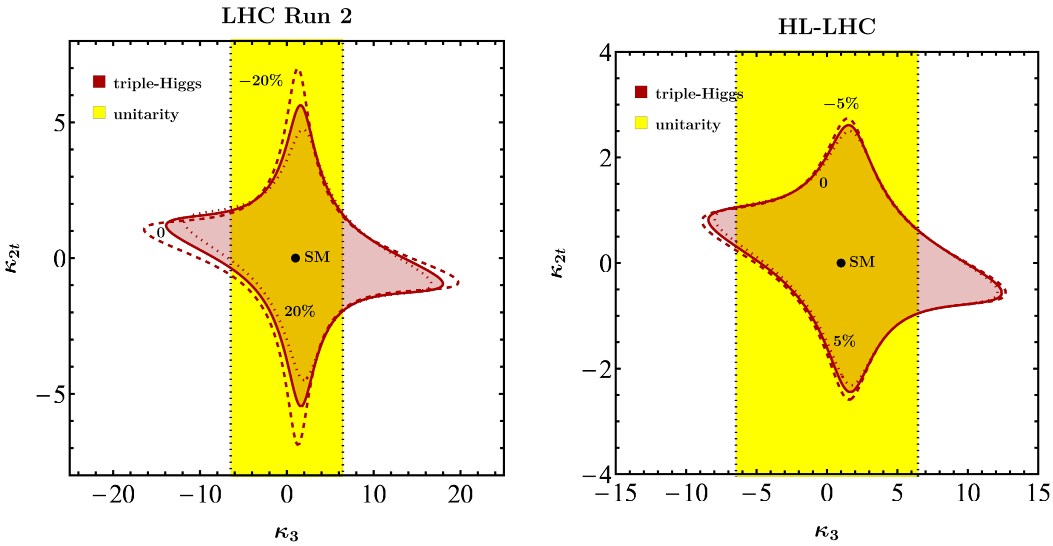

Figure 3: As Figure 2, but in the κ_3–κ_{2t} plane at LHC Run 2 (left) and the HL-LHC (right). The parameters not shown explicitly are fixed to their SM values, i.e. $\kappa_4 = 1$ and $\kappa_t = 1$.

$$c_{01}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0814\kappa_t^2 - 0.260\kappa_t^3 + 0.0866\kappa_t^4 + \kappa_{2t}\left(0.665\kappa_t - 1.10\kappa_t^2\right), \tag{21}$$

$$c_{02}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0167\kappa_t^2, \tag{22}$$

$$\begin{aligned} c_{10}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = {} & 0.204\kappa_t^2 - 1.40\kappa_t^3 + 3.56\kappa_t^4 - 3.21\kappa_t^5 \\ & + \kappa_{2t}\left(3.43\kappa_t - 13.2\kappa_t^2 + 15.3\kappa_t^3\right) + \kappa_{2t}^2\left(13.8 - 22.1\kappa_t\right) \end{aligned} \tag{23}$$

$$c_{11}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0961\kappa_t^2 - 0.260\kappa_t^3 + 0.665\kappa_{2t}\kappa_t, \tag{24}$$

$$c_{20}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.283\kappa_t^2 - 1.23\kappa_t^3 + 1.81\kappa_t^4 + \kappa_{2t}\left(2.77\kappa_t - 6.59\kappa_t^2\right) + 6.89\kappa_{2t}^2, \tag{25}$$

$$c_{21}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0480\kappa_t^2 \tag{26}$$

$$c_{30}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.155\kappa_t^2 - 0.408\kappa_t^3 + 0.922\kappa_{2t}\kappa_t, \tag{27}$$

$$c_{40}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0388\kappa_t^2. \tag{28}$$

Finally, for $\sqrt{s} = 14\,\mathrm{TeV}$, we obtain

$$\begin{aligned} c_{00}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = {} & -2.18\cdot 10^{-3} + 0.0323\kappa_t^2 - 0.311\kappa_t^3 + 1.35\kappa_t^4 - 2.92\kappa_t^5 + 2.84\kappa_t^6 \\ & + \kappa_{2t}\left(1.61\kappa_t^2 - 7.79\kappa_t^3 + 15.5\kappa_t^4 - 14.7\kappa_t^5\right) \\ & + \kappa_{2t}^2\left(6.99 - 22.4\kappa_t + 37.4\kappa_t^2\right), \end{aligned} \tag{29}$$

$$c_{01}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0822\kappa_t^2 - 0.262\kappa_t^3 + 0.0865\kappa_t^4 + \kappa_{2t}\left(0.672\kappa_t - 1.12\kappa_t^2\right), \tag{30}$$

$$c_{02}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0169\kappa_t^2, \tag{31}$$

$$\begin{aligned} c_{10}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = {} & 0.194\kappa_t^2 - 1.39\kappa_t^3 + 3.58\kappa_t^4 - 3.24\kappa_t^5 \\ & + \kappa_{2t}\left(3.46\kappa_t - 13.3\kappa_t^2 + 15.5\kappa_t^3\right) + \kappa_{2t}^2\left(14.0 - 22.4\kappa_t\right) \end{aligned} \tag{32}$$

$$c_{11}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0968\kappa_t^2 - 0.262\kappa_t^3 + 0.672\kappa_{2t}\kappa_t, \tag{33}$$

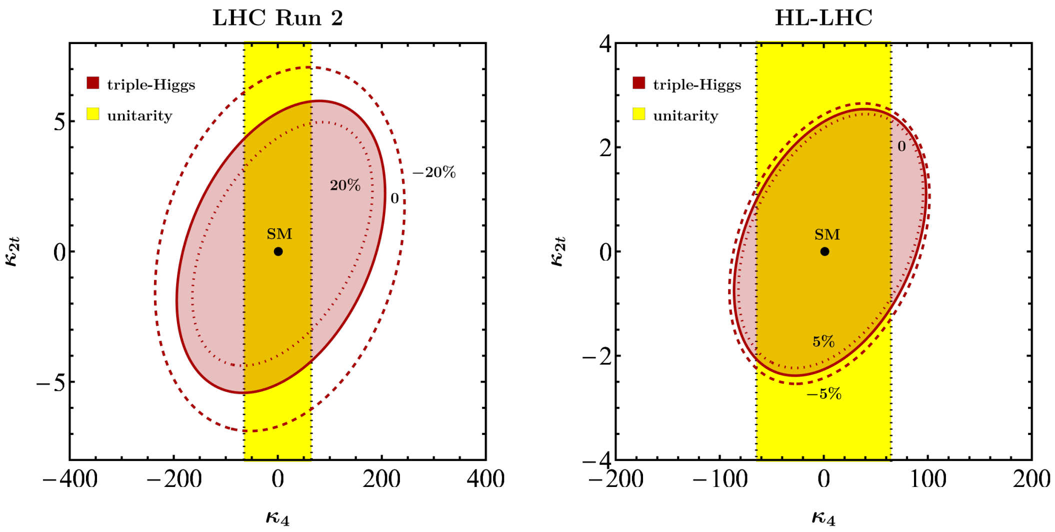

Figure 4: As Figure 2, but in the κ_4–κ_{2t} plane at LHC Run 2 (left) and the HL-LHC (right). The remaining parameters are fixed to their SM values, i.e. $\kappa_3 = 1$ and $\kappa_t = 1$.

$$c_{20}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.284\kappa_t^2 - 1.23\kappa_t^3 + 1.82\kappa_t^4 + \kappa_{2t}\left(2.79\kappa_t - 6.65\kappa_t^2\right) + 6.99\kappa_{2t}^2\,, \tag{34}$$

$$c_{21}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0484\kappa_t^2 \tag{35}$$

$$c_{30}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.156\kappa_t^2 - 0.411\kappa_t^3 + 0.931\kappa_{2t}\kappa_t\,, \tag{36}$$

$$c_{40}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0391\kappa_t^2\,. \tag{37}$$

Notice that the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ depend only mildly on the collision energy over the range considered. Likewise, the choice of PDFs is expected to have a negligible numerical impact on the parameterisations given in Eqs. (11) to (37). Higher-order QCD corrections to the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ are currently unknown, although they could, in principle, be estimated in the infinite top-quark mass limit. In the SM, these corrections increase the inclusive cross section by a K-factor of approximately 1.7 at NNLO in QCD [18]. Parameterisations of the LHC triple-Higgs gluon-fusion production cross section similar to Eqs. (9) to (37) have also been presented in Refs. [35, 36, 38, 39].

The two panels in Figure 2 illustrate the expected sensitivity to the Higgs self-couplings κ_3 and κ_4 using current LHC Run 2 data and HL-LHC projections, assuming an integrated luminosity of $3\,\mathrm{ab}^{-1}$. The red contours indicate the 95% CL constraints derived from triple-Higgs production in gluon fusion using the parameterisations in Eqs. (11) to (19) and Eqs. (29) to (37), with the normalisation to the corresponding LO SM cross section given in Eq. (10). For LHC Run 2, the triple-Higgs constraints are obtained by imposing $\mu_{3h}^{\text{LHC Run 2}} < 588$ [20], while for the HL-LHC we assume a projected upper limit of $\mu_{3h}^{\text{HL-LHC}} < 125$ [47]. Throughout this analysis, we set $\kappa_{2t} = 0$, and the black dots indicate the SM prediction. The left panel demonstrates that modifications of the top-quark Yukawa coupling of $\Delta\kappa_t = \pm 20\%$, where $\Delta\kappa_t = \kappa_t - 1$, which remain allowed at the 95% CL after LHC Run 2 [48, 49], have a significant impact on the resulting constraints. This behaviour was already pointed out in Ref. [41] and originates from the strong power-law dependence of the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ appearing in Eqs. (11) to (37) on κ_t. By the end of the HL-LHC, the top-quark Yukawa coupling is expected to be measured with a precision of $\Delta\kappa_t = \pm 5\%$ at the 95% CL [50]. Consequently, its impact on the $gg \to 3h$ constraints in the κ_3–κ_4 plane will be substantially reduced, as illustrated in the right panel. Finally, we note that the projected HL-LHC constraints from $gg \to 3h$

are comparable to those imposed by perturbative unitarity [34, 51, 52], shown as the yellow region enclosed by the black-dotted contour in both panels.

The panels in Figures 3 and 4 show the expected sensitivity in the κ_3–κ_{2t} and κ_4–κ_{2t} planes, respectively. The red contours represent the 95% CL constraints from triple-Higgs production in gluon fusion, obtained following the procedure described above. Parameters not shown explicitly are fixed to their SM values, and the black dots indicate the SM prediction. As in Figure 2, the left panels demonstrate that variations of the top-quark Yukawa coupling of $\Delta\kappa_t = \pm 20\%$ can have a significant impact on the resulting constraints, whereas the smaller variations $\Delta\kappa_t = \pm 5\%$ considered in the right panels lead to only minor changes. For κ_{2t}, triple-Higgs production at LHC Run 2 yields $-5.1 < \kappa_{2t} < 5.3$, while the HL-LHC is expected to improve this range to $-2.3 < \kappa_{2t} < 2.5$ at 95% CL. These limits are obtained by fixing $\kappa_3 = \kappa_4 = \kappa_t = 1$. We note that, under the same assumptions, the 95% CL constraint on the double-Higgs production signal strength obtained at LHC Run 2, $\mu_{2h}^{\text{LHC Run 2}} < 2.5$ [17], already implies $-0.19 < \kappa_{2t} < 0.72$. The quoted bound is obtained using the expressions given in Refs. [53, 54]. Consequently, triple-Higgs production is not expected to provide a stronger constraint on κ_{2t} than double-Higgs production in a single-parameter analysis. It can, however, provide complementary information in multi-parameter fits, in particular when these involve κ_4, which can only be weakly constrained by double-Higgs production [21–24].

2.1 Including the two-top–three-Higgs interaction

We next extend the inclusive-rate parametrisation to the contact interaction between two top quarks and three Higgs bosons in Eq. (8). We generate loop-induced LO parton-level samples with `MadGraph5_aMC@NLO` 3.5.16 [55] and a restricted UFO implementation of the κ_{3t} contact term. We again use $m_h = 125\,\text{GeV}$, $m_t = 173\,\text{GeV}$, the central member of `PDF4LHC21_40` (LHAPDF ID 93100), and $\mu_R = \mu_F = m_{3h}/2$. In this scan $\kappa_t = 1$ and $\kappa_{2t} = 0$ are fixed. The generated MadGraph amplitudes include bottom-quark loops with $m_b = 4.70\,\text{GeV}$; the 0.07 GeV difference from the MCFM input has a negligible effect at the precision of the fit.

Following the form of Eq. (9), we write the extended cross section as

$$\sigma_{\text{LO}}^{\sqrt{s}} = \left(\sigma_{\text{LO}}^{\sqrt{s}}\right)_{\text{SM}} \sum_{n,m} c_{nm}^{\sqrt{s}}(\kappa_{3t}) \left(\Delta\kappa_3\right)^n \left(\Delta\kappa_4\right)^m . \tag{38}$$

This is an exact regrouping of the LO coupling basis at the fixed values $\kappa_t = 1$ and $\kappa_{2t} = 0$. Since the amplitude is linear in κ_{3t}, c_{00} is at most quadratic in κ_{3t}, while c_{01}, c_{10}, and c_{20} are at most linear; the remaining nonzero coefficient functions are constant. The fitted SM cross sections are

$$\left(\sigma_{\text{LO}}^{13\,\text{TeV}}\right)_{\text{SM}} = 42.12\,\text{ab}\,, \qquad \left(\sigma_{\text{LO}}^{13.6\,\text{TeV}}\right)_{\text{SM}} = 47.40\,\text{ab}\,, \qquad \left(\sigma_{\text{LO}}^{14\,\text{TeV}}\right)_{\text{SM}} = 51.08\,\text{ab}\,. \tag{39}$$

They agree at the percent level with the independent values in Eq. (10).

At each collision energy, the fit uses 15 points with $\kappa_{3t} = 0$ and 20 nonzero-κ_{3t} anchor points, initially evaluated with 20 000 events per point. One anchor with a relative integration uncertainty above 0.25% was repeated with 100 000 events. Six additional off-grid points per energy are reserved for validation. The normalisation fixes $c_{00}^{\sqrt{s}}(0) = 1$. The fitted coefficient functions are

For $\sqrt{s} = 13\,\text{TeV}$, the non-vanishing coefficient functions are

$$c_{00}^{13\,\text{TeV}}(\kappa_{3t}) = 1 - 2.83183\,\kappa_{3t} + 15.81829\,\kappa_{3t}^2\,, \tag{40}$$

$$c_{01}^{13\,\text{TeV}}(\kappa_{3t}) = -0.0914560 + 0.793591\,\kappa_{3t}\,, \tag{41}$$

$$c_{02}^{13\,\text{TeV}}(\kappa_{3t}) = 0.0168979\,, \tag{42}$$

$$c_{10}^{13\,\mathrm{TeV}}(\kappa_{3t}) = -0.865038 - 3.92841\,\kappa_{3t}\,, \tag{43}$$
$$c_{11}^{13\,\mathrm{TeV}}(\kappa_{3t}) = -0.167158\,, \tag{44}$$
$$c_{20}^{13\,\mathrm{TeV}}(\kappa_{3t}) = 0.872294 + 0.900894\,\kappa_{3t}\,, \tag{45}$$
$$c_{21}^{13\,\mathrm{TeV}}(\kappa_{3t}) = 0.0487935\,, \tag{46}$$
$$c_{30}^{13\,\mathrm{TeV}}(\kappa_{3t}) = -0.259375\,, \tag{47}$$
$$c_{40}^{13\,\mathrm{TeV}}(\kappa_{3t}) = 0.0395354\,. \tag{48}$$

For $\sqrt{s} = 13.6\,\mathrm{TeV}$, they are

$$c_{00}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 1 - 2.87872\,\kappa_{3t} + 16.21306\,\kappa_{3t}^2\,, \tag{49}$$
$$c_{01}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = -0.0909154 + 0.800038\,\kappa_{3t}\,, \tag{50}$$
$$c_{02}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 0.0168638\,, \tag{51}$$
$$c_{10}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = -0.866564 - 3.95165\,\kappa_{3t}\,, \tag{52}$$
$$c_{11}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = -0.166501\,, \tag{53}$$
$$c_{20}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 0.871502 + 0.897150\,\kappa_{3t}\,, \tag{54}$$
$$c_{21}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 0.0485042\,, \tag{55}$$
$$c_{30}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = -0.257216\,, \tag{56}$$
$$c_{40}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 0.0390715\,. \tag{57}$$

For $\sqrt{s} = 14\,\mathrm{TeV}$, they are

$$c_{00}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 1 - 2.95187\,\kappa_{3t} + 16.47338\,\kappa_{3t}^2\,, \tag{58}$$
$$c_{01}^{14\,\mathrm{TeV}}(\kappa_{3t}) = -0.0921387 + 0.803365\,\kappa_{3t}\,, \tag{59}$$
$$c_{02}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 0.0168553\,, \tag{60}$$
$$c_{10}^{14\,\mathrm{TeV}}(\kappa_{3t}) = -0.853992 - 3.93891\,\kappa_{3t}\,, \tag{61}$$
$$c_{11}^{14\,\mathrm{TeV}}(\kappa_{3t}) = -0.166145\,, \tag{62}$$
$$c_{20}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 0.863440 + 0.899872\,\kappa_{3t}\,, \tag{63}$$
$$c_{21}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 0.0482948\,, \tag{64}$$
$$c_{30}^{14\,\mathrm{TeV}}(\kappa_{3t}) = -0.255934\,, \tag{65}$$
$$c_{40}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 0.0390078\,. \tag{66}$$

The 18 independent validation rates are reproduced to better than 0.9% in relative terms. The fit values of χ^2/dof are 56.5/21, 40.9/21, and 78.9/21 at 13, 13.6, and 14 TeV, respectively, indicating that the residual numerical scatter is not fully described by the quoted integration errors. In particular, the broad 14 TeV validation point $(\kappa_3, \kappa_4, \kappa_{3t}) = (-2, -25, -1.5)$ has a pull of -4.00 while its fractional residual is only -0.37%. We therefore use Eq. (38) for the phenomenological illustrations below but regard its last sub-percent digits as preliminary pending higher-statistics validation.

Figures 5 and 6 are the κ_{3t} analogues of Figures 3 and 4. Fixing all other couplings to their SM values yields

$$-6.00 < \kappa_{3t} < 6.18 \qquad (\text{LHC Run 2})\,, \qquad -2.66 < \kappa_{3t} < 2.83 \qquad (\text{HL-LHC})\,. \tag{67}$$

The quadratic form also exposes an inclusive-rate degeneracy. At $\kappa_3 = \kappa_4 = 1$, the two solutions of $c_{00}^{\sqrt{s}}(\kappa_{3t}) = 1$ are $\kappa_{3t} = 0$ and $\kappa_{3t} \simeq 0.179$ at all three energies to the accuracy of the fit. Differential information is therefore essential for distinguishing these two solutions.

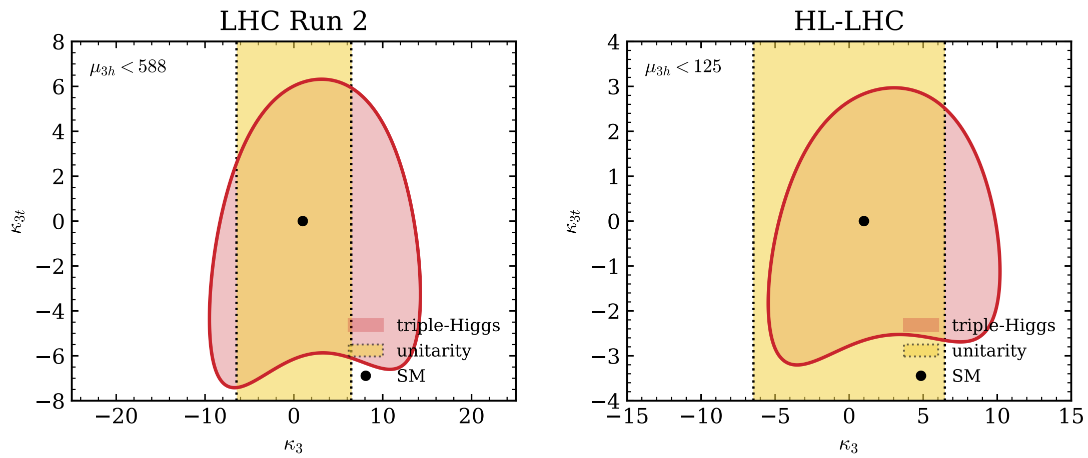

Figure 5: Constraints in the κ_3–κ_{3t} plane at LHC Run 2 (left) and the HL-LHC (right), obtained from Eq. (38) using $\mu_{3h} < 588$ and $\mu_{3h} < 125$, respectively. The red regions are allowed by the inclusive triple-Higgs rate, the black points denote the SM, and the yellow vertical bands show the projection of the tree-level $hh \to hh$ perturbative-unitarity constraint. We fix $\kappa_4 = \kappa_t = 1$ and $\kappa_{2t} = 0$. The yellow band constrains κ_3 only; no independent unitarity condition on κ_{3t} is imposed.

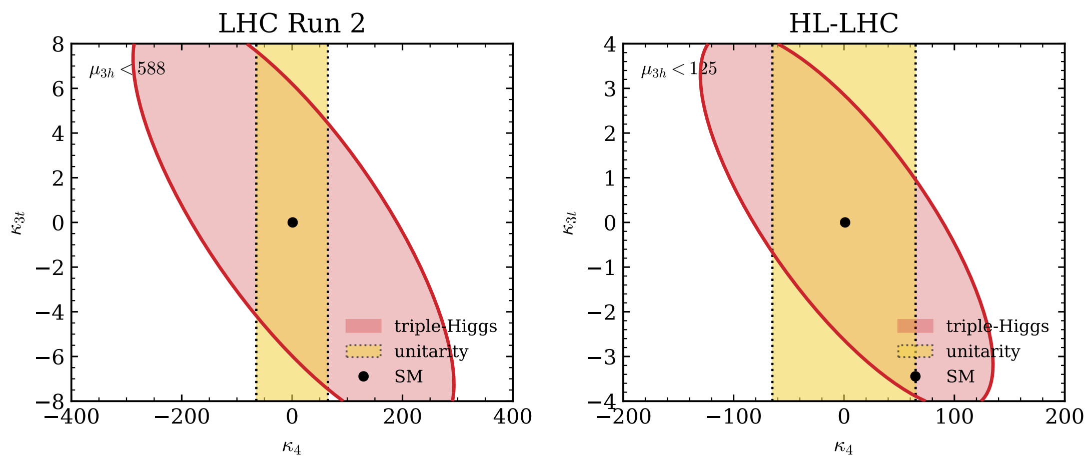

Figure 6: As in Figure 5, but in the κ_4–κ_{3t} plane with $\kappa_3 = \kappa_t = 1$ and $\kappa_{2t} = 0$. The yellow band is the projection of the $hh \to hh$ perturbative-unitarity condition onto κ_4.

Modifications of the Higgs trilinear and quartic self-couplings affect not only the inclusive triple-Higgs production cross section but also the kinematic properties of the $gg \to 3h$ final state. To illustrate this point, Figure 7 shows normalised kinematic distributions at the HL-LHC for the SM and two BSM benchmark scenarios. Both benchmark points yield the same inclusive signal strength, $\mu_{3h} = 65$, with inclusive cross sections differing by only about 5%, while remaining consistent with the perturbative unitarity bounds obtained from tree-level $hh \to hh$ scattering. The left and right panels show the invariant mass of the three-Higgs system, m_{3h}, and the scalar sum of the Higgs transverse momenta, $\sum p_{T,h}$, respectively. All results are obtained for $\kappa_t = 1$ and $\kappa_{2t} = 0$. This choice is motivated by the observation of Ref. [41] that variations of κ_t approximately induce an overall rescaling of the normalised distributions, without significantly altering their shapes. The comparison demonstrates that values of κ_3 and κ_4 yielding nearly degenerate inclusive cross sections can nevertheless lead to size-

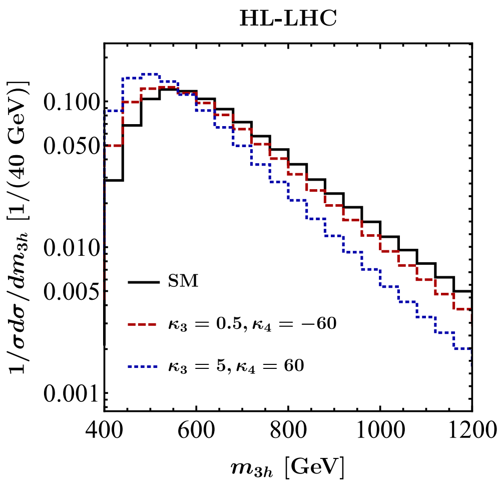

Figure 7: Normalised kinematic distributions for $gg \to 3h$ production at the HL-LHC in the SM (solid black line) and in two benchmark scenarios defined by κ_3 and κ_4 (red dashed and blue dotted lines). The left and right panels show the invariant mass of the three-Higgs system, m_{3h}, and the scalar sum of the transverse momenta of the Higgs bosons, $\sum p_{T,h}$, respectively. All results are obtained for $\kappa_t = 1$ and $\kappa_{2t} = 0$.

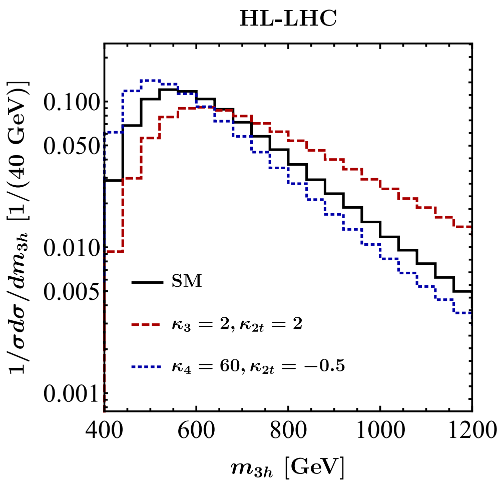

Figure 8: As in Figure 7, but for two other κ-benchmark scenarios indicated in the panels. The remaining parameters are fixed to their SM values.

able distortions of the differential spectra, reaching up to about 50% in both the m_{3h} and $\sum p_{T,h}$ distributions, in the peak regions as well as in the high-energy tails. Consequently, exploiting differential information in future HL-LHC analyses of $gg \to 3h$ production has the potential to improve the constraint quoted below Eq. (6), which currently relies on the inclusive rate only [21].

Figure 8 presents additional normalised kinematic distributions at the HL-LHC for the SM and two further BSM benchmark scenarios within the κ-framework. The benchmark points are selected such that both lead to an enhanced signal strength of $\mu_{3h} = 75$, while their inclusive cross sections differ by less than about 5%. The corresponding values of κ_3 and κ_4 are consistent with perturbative unitarity constraints, and all remaining parameters are fixed to their SM values. As in Figure 7, sizeable distortions of the differential spectra arise despite the near-degeneracy of the inclusive rates. This further demonstrates that exploiting differ-

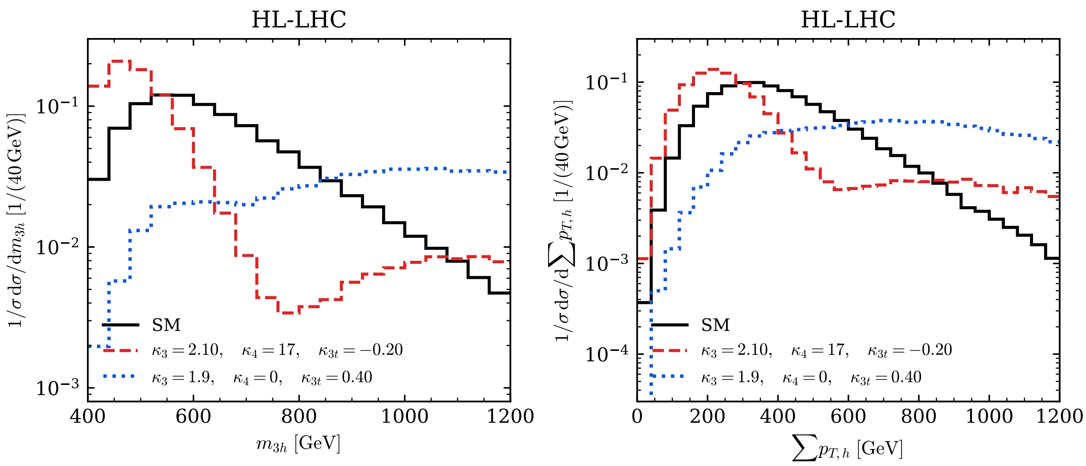

Figure 9: Normalised m_{3h} and $\sum p_{T,h}$ distributions at the HL-LHC for the SM and the two rate-matched benchmarks $(\kappa_3, \kappa_4, \kappa_{3t}) = (2.10, 17, -0.20)$ and $(1.9, 0, 0.40)$. We fix $\kappa_t = 1$ and $\kappa_{2t} = 0$. Each curve contains 100 000 generated events.

ential information in future HL-LHC searches for $gg \to 3h$ production can help lift parameter degeneracies within the κ-framework.

The two mixed-coupling benchmarks in Figure 9 provide more dramatic examples. Their inclusive cross sections are 43.571 ab and 41.744 ab, respectively, compared with 43.391 ab for the SM sample, corresponding to rate ratios of 1.004 and 0.962. Nevertheless, the normalised bin-by-bin ratios exhibit order-one changes. For $(2.10, 17, -0.20)$ they range from approximately 0.07 to 4.6 in m_{3h} and from 0.17 to 3.7 in $\sum p_{T,h}$; for $(1.9, 0, 0.40)$ the corresponding ranges are approximately 0.07–7.2 and 0.10–12.7. These ranges retain only bins containing at least 0.2% of the normalised SM sample, reducing sensitivity to statistical fluctuations in the far tails. The pronounced dip of the first benchmark around $m_{3h} \simeq 750$–800 GeV is generated by destructive interference between the standard loop topologies and the amplitudes involving the modified self-couplings and κ_{3t} contact interaction; compensating enhancements at lower invariant mass and in parts of the tail keep the inclusive rate SM-like.

The dedicated distribution campaign instead uses the central member of NNPDF40_lo_as_01180 (LHAPDF ID 331900). MadGraph dynamical-scale choice 3 sets

$$\mu_R = \mu_F = \frac{H_T}{2}, \qquad H_T = \sum_h \sqrt{m_h^2 + p_{T,h}^2}. \tag{68}$$

The quoted rates use their common 43.391 ab SM reference rather than Eq. (39), which employs the PDF4LHC21_40 and $m_{3h}/2$ setup. The rate ratios and normalised shape comparisons in Figure 9 are internally consistent.

3 Conclusions

Unlike the Higgs couplings to EW gauge bosons and third-generation fermions, which are already being measured with increasing precision, the Higgs self-interactions remain largely unconstrained by LHC Run 2 data. This leaves considerable room for BSM scenarios that modify the shape of the Higgs potential. Indeed, both theoretical considerations and explicit ultraviolet-complete constructions demonstrate that deviations of the Higgs self-couplings exceeding 200% can be consistent with existing single-Higgs measurements without requiring fine-tuned parameter choices [4]. Measurements of multi-Higgs production at the HL-LHC and future colliders therefore provide a unique opportunity to probe largely unexplored regions of parameter space and retain significant discovery potential, even in scenarios where single-Higgs observables remain close to their SM predictions.

In this note, we have derived compact "pocket formulas" for triple-Higgs production within the κ-framework, enabling constraints on the Higgs trilinear and quartic self-couplings to be obtained from current and future measurements of the $gg \to 3h$ signal strength. The original parameterisation retains the full dependence on κ_3, κ_4, κ_t, and κ_{2t} at LO in perturbation theory. We have supplemented it with Eq. (38), which includes the complete LO dependence on κ_{3t} for $\kappa_t = 1$ and $\kappa_{2t} = 0$ at 13, 13.6, and 14 TeV. Since the derivation relies on flexible matrix-element implementations, further extensions to incorporate additional coupling modifications — such as those arising at next-to-leading order in the Higgs Effective Field Theory [54] — are comparatively straightforward.

Using these parameterisations, we have investigated the sensitivity to κ_3 and κ_4 using current and projected LHC data. We find that the presently allowed uncertainty in κ_t can have a substantial impact on the constraints in the κ_3–κ_4 plane. This dependence is significantly reduced at the HL-LHC, where the expected percent-level precision of the top-quark Yukawa coupling measurement will strongly restrict the corresponding parameter space. For κ_{2t}, which parametrises the interaction between two top quarks and two Higgs bosons, we find that triple-Higgs production is unlikely to provide stronger bounds than double-Higgs production in a single-parameter analysis. Nevertheless, it can yield complementary information in global multi-parameter fits, especially once κ_4 is included, as this coupling remains only weakly constrained by double-Higgs measurements [21–24]. In a single-parameter interpretation, the present and projected triple-Higgs rate assumptions give $-6.00 < \kappa_{3t} < 6.18$ and $-2.66 < \kappa_{3t} < 2.83$, respectively. The nonzero solution near $\kappa_{3t} = 0.179$ is rate-degenerate with the SM. We further demonstrate with 100 000-event samples that this degeneracy is lifted by the m_{3h} and $\sum p_{T,h}$ shapes, and identify mixed $(\kappa_3, \kappa_4, \kappa_{3t})$ benchmarks whose inclusive rates agree with the SM within 4% while their binned spectra differ by factors of several.

The projected HL-LHC sensitivity to κ_4 in a single-parameter analysis remains modest, with constraints reaching only $|\kappa_4| \lesssim 80$, comparable to those derived from perturbative unitarity arguments [34, 51, 52]. An improvement by approximately one order of magnitude in the determination of the Higgs quartic self-coupling could be achieved at a hadron-collider option of the Future Circular Collider [22, 28–30, 32, 33, 35–38, 40]. Reaching a precision at the level of 50%, however, would likely require the capabilities of a future muon collider [56].

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